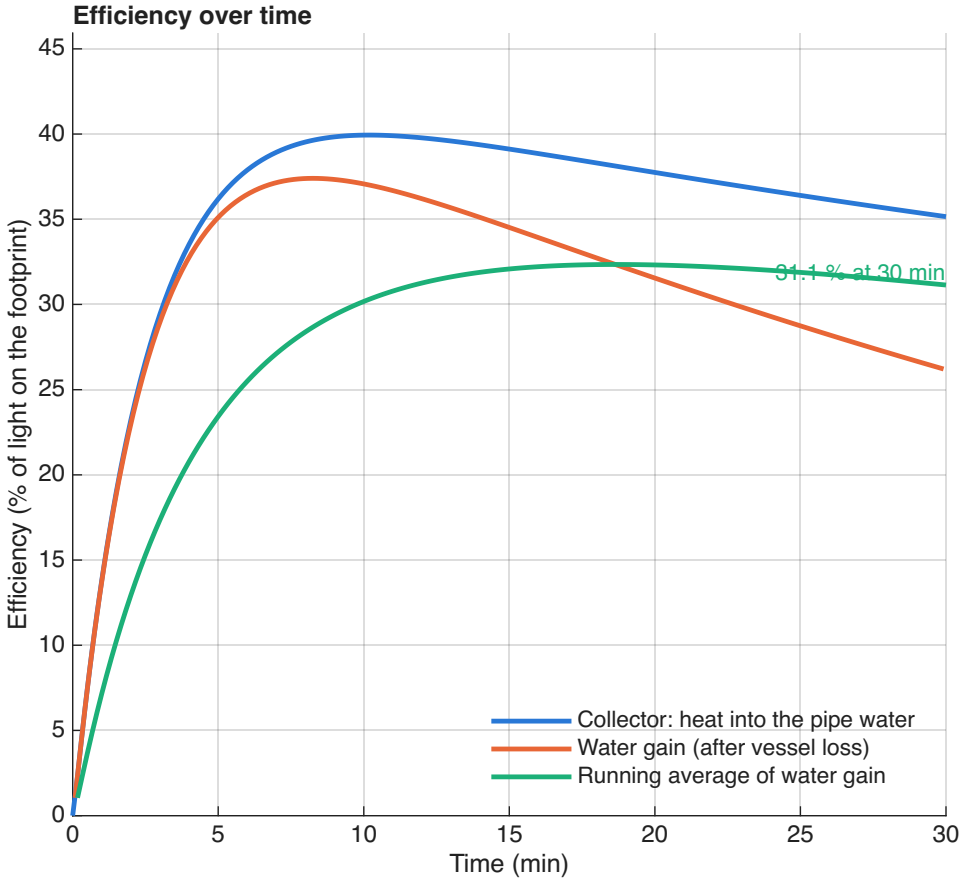
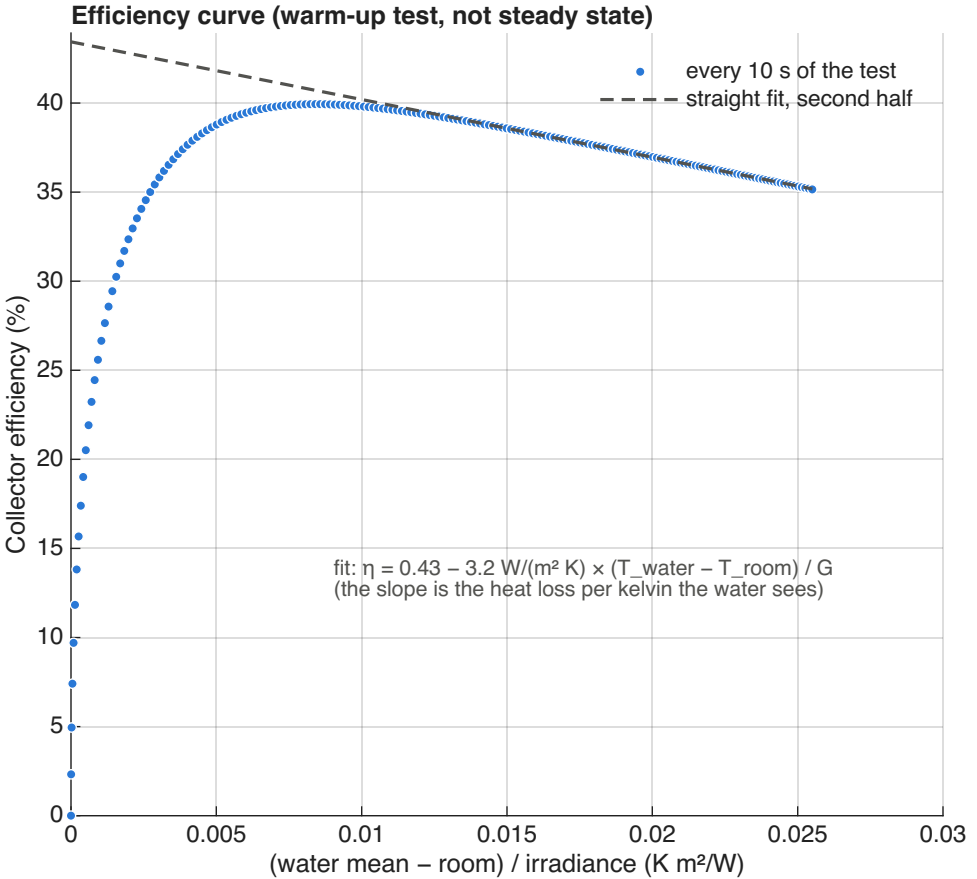

Efficiency over time
Efficiency (% of light on the footprint)
Time (min)
31.1 % at 30 min
Collector: heat into the pipe water
Water gain (after vessel loss)
Running average of water gain


Efficiency curve (warm-up test, not steady state)
Collector efficiency (%)
(water mean – room) / irradiance (K m²/W)
every 10 s of the test
straight fit, second half
fit: η = 0.43 – 3.2 W/(m² K) × (T_water – T_room) / G
(the slope is the heat loss per kelvin the water sees)